Amazon Fresh Sales Analysis & Business Insights Dashboard Using SQL & Power BI

SQL Mini Project Report

Submitted by: Mohana Shanthini.S

Course: Data Analytics

Institute: Vinsup Skill Academy

1.Introduction:

Sales and customer data play an important role in helping businesses understand their performance and make informed decisions. This project, "Amazon Fresh Sales Analysis and Business Insights Dashboard Using SQL and Power BI," was carried out to apply SQL techniques for data management and analysis and Power BI for data visualization and reporting. The project covers the complete workflow from data modeling, data preparation, and SQL querying to interactive dashboard development. It provides practical experience in analyzing sales, customer behaviour, product performance, inventory, and regional trends. The dashboards present key business metrics and meaningful insights that support effective decision-making and improve overall business performance.

2.Problem Statement:

Amazon Fresh maintains a large volume of data related to customers, products, orders, inventory, and sales across different product categories and regions. Since the data is stored across multiple tables in raw format, it is difficult to retrieve meaningful information, identify sales trends, compare regional performance, and evaluate customer purchasing behaviour efficiently. To overcome this challenge, SQL was used to manage, retrieve, clean, and analyze the data through queries, while Power BI was used to transform the analyzed data into interactive dashboards. The dashboards present key business metrics and visual insights, enabling easier analysis, data-driven decision-making, and improved business performance.

3.Objectives:

The main objectives of this project were:

- To design and implement the Amazon Fresh database using SQL for efficient data management.
- To clean, retrieve and analyze business data using SQL queries and analytical functions.
- To create data models and establish relationships between multiple tables for accurate analysis.
- To analyze sales performance, product inventory, customer demographics and regional sales trends.

- To develop interactive Power BI dashboards with KPIs, charts, slicers and navigation features for effective business reporting.
- To generate meaningful business insights and recommendations that support data-driven decision-making and improve overall business performance.

4.Dataset Description:

The project uses an **Amazon Fresh sales dataset** consisting of multiple related tables, including Customers, Products, Orders, Order Details, Reviews, Suppliers, Categories, and States. The dataset contains information about sales transactions, customer details, product information, and order records. SQL was used to manage and analyze the data by establishing relationships between the tables. Power BI was used to visualize the analyzed data through interactive dashboards. This dataset supports comprehensive business analysis and data-driven decision-making.

5.Software and Technologies Used

- **Microsoft Excel**
    - Used for initial data review and validation.
    - Verified dataset structure and checked for inconsistencies before database implementation.
- **MySQL Workbench**
    - Used to create and manage the relational database.
    - Designed database tables and established primary and foreign key relationships.
    - Executed SQL queries for data retrieval, analysis, and business insights.
- **Microsoft Power BI**
    - Connected with the SQL database for data visualization.
    - Performed data transformation using Power Query.
    - Created DAX measures, KPI cards, interactive charts, slicers, and dashboards.
    - Generated visual reports to support business analysis and decision-making.

6.Project Workflow

The project followed a structured workflow, starting from database planning and progressing through SQL analysis to interactive dashboard development in Microsoft Power BI.

1. Database Planning

- Studied the dataset and identified the required business entities.
- Designed the Entity Relationship (ER) Diagram to represent table relationships.
- Defined primary keys and foreign keys to establish a relational database structure.
- Ensured data integrity and efficient data organization.

2. Database Development

- Created database tables using SQL Data Definition Language (DDL) commands.
- Implemented database constraints to maintain data consistency.
- Performed data insertion, updating, and maintenance using SQL Data Manipulation Language (DML) commands.
- Verified the accuracy of stored data before analysis.

3. Data Analysis

- Executed SQL queries to retrieve and analyze business information.
- Applied JOIN operations to combine data from multiple tables.
- Used aggregate functions, GROUP BY, HAVING, ORDER BY, and ranking functions to generate business insights.
- Evaluated sales performance, customer behaviour, product performance, and regional trends.

4. Data Preparation for Visualization

- Connected the SQL database to Microsoft Power BI.
- Used Power Query to clean, transform, and prepare the dataset.
- Established relationships between tables to build an optimized data model.

5. Dashboard Development

- Designed interactive dashboards using KPI cards, charts, slicers, and filters.
- Developed DAX measures to calculate business metrics.
- Organized dashboards into multiple analytical pages for better user experience.
- Enabled interactive filtering and navigation for effective business analysis.

7. Project Implementation

The implementation of the Amazon Fresh Sales Analysis project involved transforming raw business data into a structured analytical solution. The project combined database technologies and business intelligence tools to extract meaningful insights and present them through interactive dashboards.

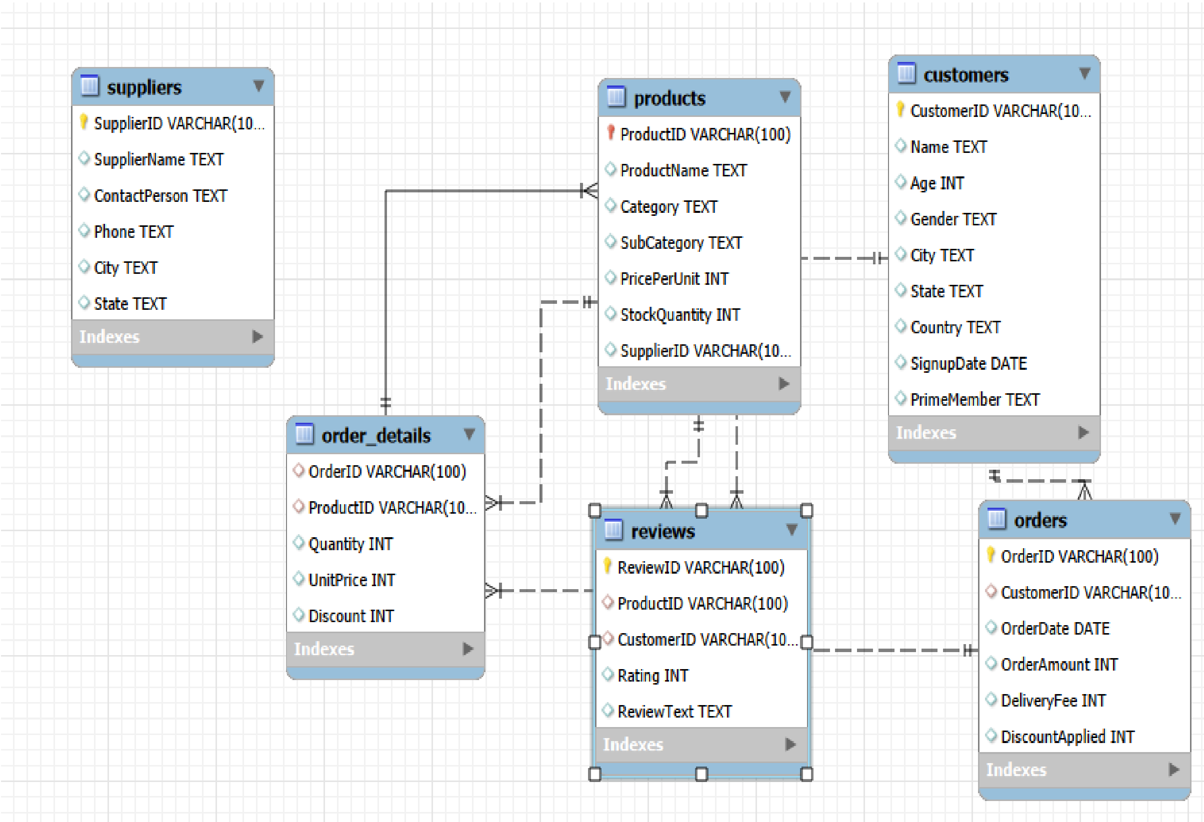
- **Dataset Examination**

The project began with a detailed examination of the Amazon Fresh dataset using Microsoft Excel. The dataset was inspected to understand the available business information, identify relationships among tables, and verify the completeness of records. Initial validation was carried out to detect duplicate entries, missing values, and formatting inconsistencies before importing the data into the database.

7.2 Relational Database Development

- **Database Schema Design**

A relational database was developed in MySQL Workbench by organizing the dataset into multiple business entities, including Customers, Products, Categories, Suppliers, Orders, Order Details, Reviews, and States. An Entity Relationship (ER) Diagram was designed to define the logical connections between the tables and ensure an efficient database structure.



- **Database Construction**

The database structure was created using SQL Data Definition Language (DDL) statements. Appropriate constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK, and DEFAULT were implemented to maintain data accuracy and enforce referential integrity throughout the database.

- **Data Maintenance**

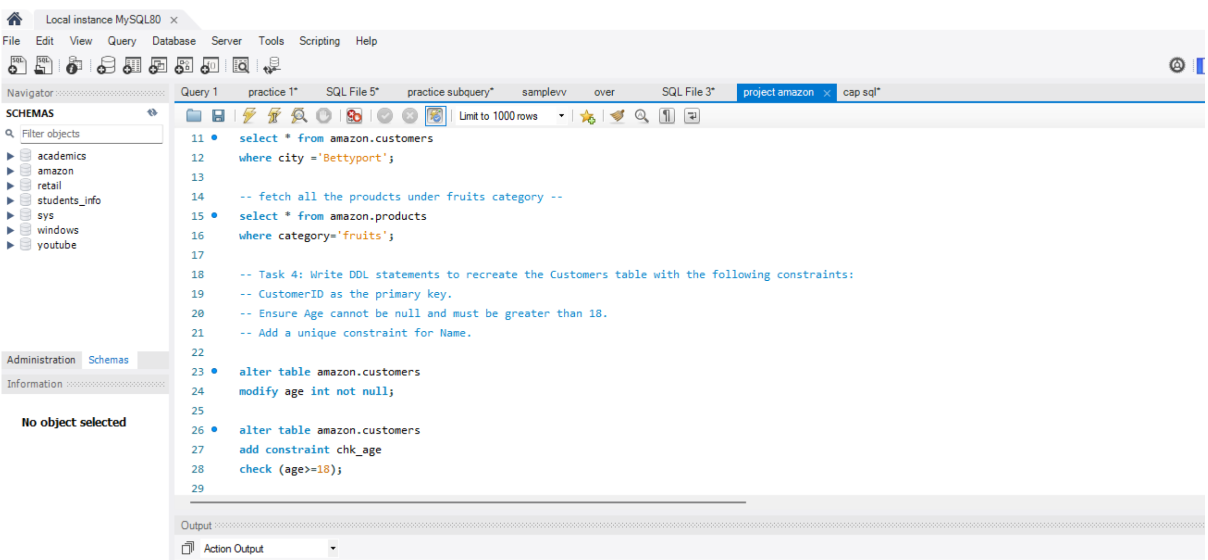
After creating the database, SQL Data Manipulation Language (DML) statements were used to manage the data. Operations such as INSERT, UPDATE, and DELETE ensured that the database remained accurate, consistent, and ready for analytical processing.

7.3 SQL-Based Business Analytics

The relational database was analyzed using SQL to answer various business questions and evaluate organizational performance.

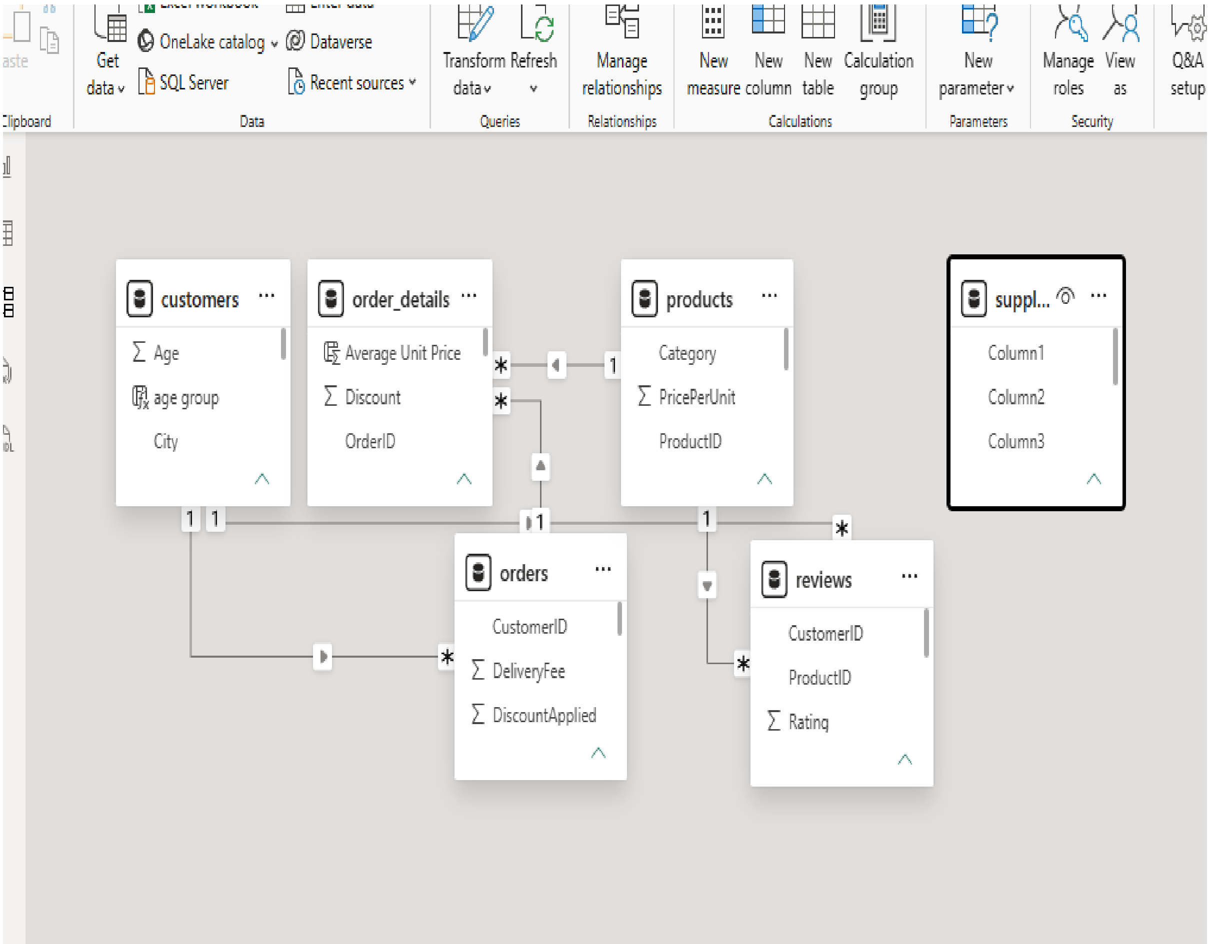
The analysis included:

- Retrieving records based on business requirements.
- Combining information from multiple tables using JOIN operations.
- Summarizing business data using GROUP BY and HAVING clauses.
- Performing calculations using SUM(), COUNT(), AVG(), MIN(), and MAX().
- Identifying top-performing customers and products using ranking techniques.
- Evaluating sales performance across different categories and states.
- Measuring customer purchasing behaviour and revenue contribution.



7.4 Data Modeling in Power BI

The processed SQL data was imported into Microsoft Power BI for reporting and visualization. Power Query was used to clean and transform the dataset before creating relationships between tables. A well-structured data model was developed to improve reporting accuracy and dashboard performance.



7.5 Dashboard Development

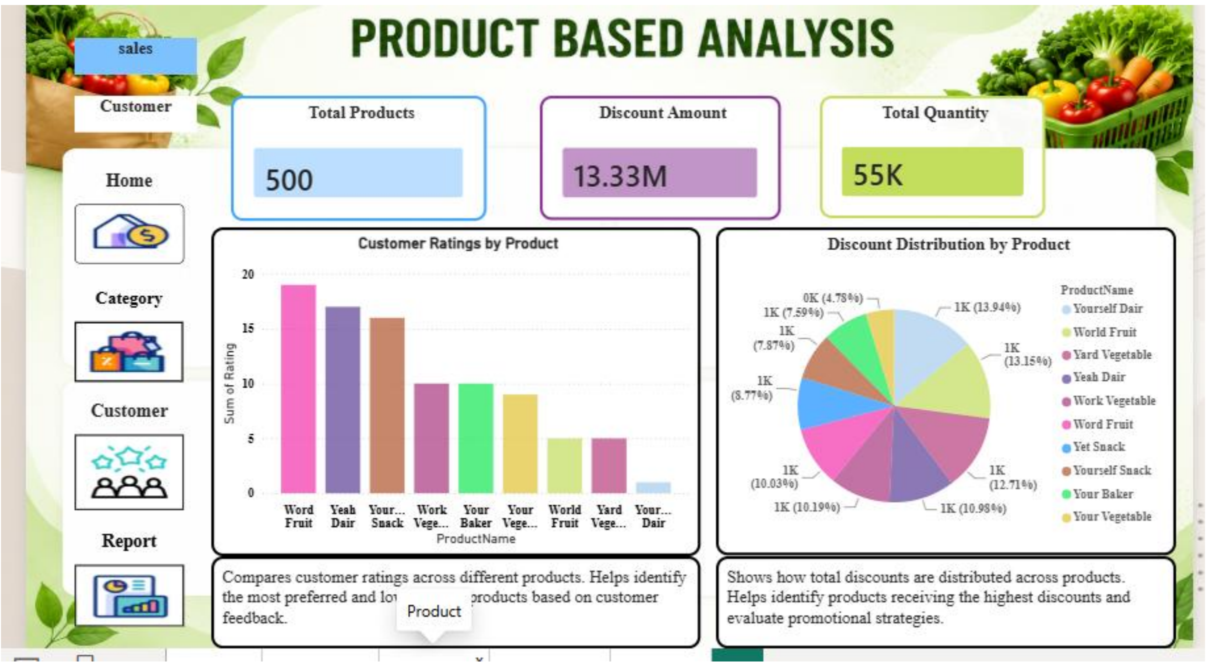
Interactive dashboards were designed to present business information in a clear and user-friendly format. Multiple report pages were developed, each focusing on a different area of business analysis.

The dashboard includes:

- Home Dashboard
- Category-wise Discount Analysis
- Product-Based Analysis

- Customer Demographics & Purchasing Behaviour
- Regional Sales & Customer Satisfaction

Each page was designed with KPI cards, charts, slicers, filters, and navigation buttons to improve user interaction and analytical efficiency.

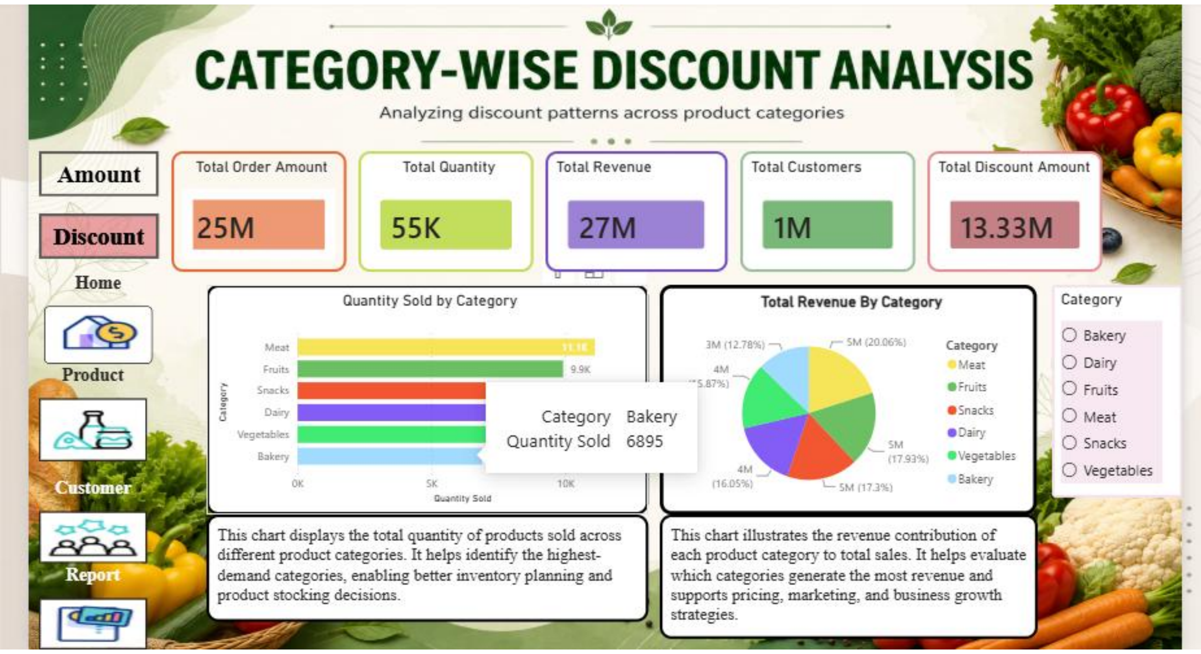


7.6 Dashboard Functionality

The dashboards include several interactive features that enhance business analysis and reporting.

- Dynamic KPI Cards
- Slicers for filtering data
- Cross-filtering between visuals
- Navigation buttons
- Drill-down analysis
- DAX measures for business calculations
- Consistent dashboard theme and formatting

These features enable users to explore business data from different perspectives and quickly identify important trends and performance indicators.



7.7 Business Outcome

The completed solution provides a comprehensive view of Amazon Fresh's business performance by integrating SQL analysis with Power BI visualizations. The dashboards enable users to monitor sales performance, evaluate product demand, analyze customer behaviour, compare regional performance, and track inventory status. The project demonstrates how database management and business intelligence tools can be combined to support informed decision-making and improve overall business efficiency.

8.Business Insights

This section will be written directly from your dashboard findings, for example:

- Meat generates the highest revenue.
- Bakery has the lowest sales.
- Texas contributes the highest revenue.
- Customer ratings are strongest for selected fruit and dairy products.
- Customers aged above 50 contribute the highest revenue.
- Current sales have reached 50% of the target.
- High discounts drive sales in selected categories.
- Inventory levels indicate which products require restocking.

9.Business Recommendations

Recommendations will be based on your analysis, such as:

- Increase inventory for high-demand products.
- Optimize discounts for low-performing categories.
- Expand successful regional sales strategies.
- Promote highly rated products.
- Improve marketing for younger age groups.
- Focus on reaching the remaining sales target through seasonal campaigns.

10.Conclusion:

The Amazon Fresh Sales Analysis Dashboard provides insights into sales, product performance, customer behaviour, inventory, and regional trends. The analysis highlights approximately ₹27 million in revenue, ₹25 million in order value, and 55,000 units sold, showing strong business performance. SQL was used for data management and analysis, while Power BI enabled interactive visualization and reporting. The dashboard helps identify business opportunities, improve operational efficiency, and support data-driven decision-making.